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Microeconomic Consequences and Macroeconomic Causes of Foreign Direct Investment in Southern African Economies¹

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Abstract. The authors use a new data set on firms in thirteen countries of the Southern African Development Community (SADC) and comparators from other regions to identify the benefits and determinants of FDI in this region. FDI has facilitated local development in the SADC. Foreign owned firms perform better than domestic firms, are larger, and locate in richer and better-governed countries, and in countries with more competitive financial intermediaries. They are also more likely to export than domestic firms and evidence suggests that they might have positive spillover effects on domestic firms. Based on a standard empirical model, the SADC is attracting the inward FDI per capita the region's level of income would predict. But this means that there are less capital inflows per capita to the region than there are to wealthier parts of the developing world. Moreover, the SADC is attracting less FDI than comparators for reasons that are possibly more fundamental than current income, namely, countries' past growth record, demographic structure and the quality of physical infrastructure. Interestingly, inward FDI is less sensitive to variation in income within the SADC than in other parts of the world, but is more responsive to changes in country's openness to trade.

Key words: FDI, spillovers, firm performance, exports, Africa, SADC.

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1. Introduction

This article offers empirical evidence on the microeconomic consequences and macroeconomic causes of Foreign Direct Investment (FDI) in the Southern African Development Community (SADC, hereafter), which is a regional economic community of 13 member states of a combined population of 250 million.² Would countries in the region grow faster if there were more FDI than is the case presently? Is the region attracting as much foreign capital as it would be expected given the region's macroeconomic performance? If it is not, what could be the reasons for it? These questions cannot be answered by referring to current literature. This article addresses these questions with analyses of the World Bank's Enterprise Surveys, a global cross country dataset that combines observations on the performance of firms with information on their ownership and overall business environment, and macroeconomic data.

A common hypothesis about the determinants and consequences of FDI inflows is that FDI can reduce income gaps between developing and advanced economies. In the neoclassical world of perfect factor mobility and technology transfer, capital readily flows from rich to poor countries, seeking higher returns in capital-scarce economies. However, we do not always see this in actual data, some of which actually suggest that hopes that FDI inflows are a "supply side remedy" to catapult poor countries into the fast track of development, as prescribed by the United Nations (1999), could be too optimistic.

This observation applies to the data being analyzed in this paper. Even though the average income of SADC countries is lower than the average of all countries covered by our data, per capita FDI inflows to the SADC region are only 36.6 U.S. dollars (in 2000 prices) per annum whereas the average for all other countries is 202.8 dollars. SADC countries are also attracting less FDI than other countries with comparable incomes, for which FDI inflows per capita average 63.2 dollars. This positive (and counter intuitive) association between per capita income and net FDI inflows is also evident within the SADC, where cross country gaps in flows are large. Within the SADC, annual per capita FDI inflows are less than 10 dollars among the low-income countries, but range between

² The members are: Angola, Botswana, the Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, the Seychelles, South Africa, Swaziland, Tanzania, Zambia, and Zimbabwe.

50 and 167 dollars per capita among the middle income countries of Angola, Botswana and South Africa.³

Turning to the consequences of FDI, another common hypothesis is that FDI can foster local economic development either because it relaxes capital constraints to growth or because it facilitates the transfer of new technology or knowhow. However, it is also plausible that foreign-owned companies merely capture rents, crowd out domestic investment, and do not provide technological spillovers for domestic firms. Even if FDI does facilitate local growth by relaxing capital constraints or through technology transfer or spillovers, the magnitude of these benefits would be hard to identify *a priori* in any setting, because they could vary across countries. For example, the benefits may depend on the market size of the host country and the time-horizon of the commitments involved in the investment. This combination of high hopes and unclear evidence puts empirical research in center stage. Thus, it is useful to investigate empirically whether FDI plays a positive role in SADC, and if so, how. Many also presume that the FDI inflow levels to the region are too low. How much is too little? What factors can explain the observed FDI inflows?

On the question of the role of FDI in the SADC, our data suggest that this is likely to have been one of facilitating growth in the region. In the data, foreign owned firms grow faster on average than domestic firms. Also, foreign owned firms are, on average, more productive, larger, and more likely to export. Furthermore, there is some evidence that domestic firms perform better in the presence of foreign owned firms than they do otherwise. At the same time, foreign owned firms are more likely to locate in richer and better-governed countries, and countries that have more competitive financial industries.

To assess the determinants of inward FDI to the region, we use an accepted empirical model described in Fan et al. (2009) in which FDI per capita is a function of market size, infrastructure, education, exchange rates, the quality of institutions, and past economic performance (as measured by the mean and variability of growth). The data are broadly consistent with the predictions of the model. However, market size appears to

³ It is worth noting that there is a similar if not more pronounced variation in FDI inflow per capita among regions within China (Wang, Xu and Zhu 2012).

be a weaker determinant of FDI in the SADC than in other countries, while openness to trade is more important than in the other countries.⁴

The next section presents our firm-level evidence on the consequences of FDI. Section three examines the macroeconomic determinants of FDI inflows with panel data. Section four concludes.

2. Micro Consequences of Foreign Ownership in SADC

If FDI supports economic development, its direct effects should show up in the growth and productivity of firms benefitting from foreign capital investment, or at least on some other indicator of performance, such as export orientation. In addition, FDI can have indirect effects on the economic performance of other domestic firms through spillover effects, as mentioned above. The mechanism by which FDI boosts productivity and growth of recipient firms could be that it addresses or relaxes the capital shortages and financial constraints that limit the scale of domestic firms. To the extent that this is true, part of the productivity advantage of foreign owned firms should come from their operating on a larger scale than domestic firms. A related but distinct source of the potential productivity advantage of foreign owned firms is that they often have better access to new technology or can afford to engage in R& D investment. If true, FDI could influence the growth and productivity of the domestic industry by becoming sources of learning spillovers for domestic firms.

To investigate the effects of FDI through these microeconomic channels, we analyze the correlation between the foreign ownership of firms and the performance of SADC firms in terms of sales growth and total factor productivity. The literature has not produced a consensus on the effect of foreign ownership on productivity (Caves, 1999).

⁴ The result relating to openness is consistent with the view that FDI tends to be more export oriented in smaller countries. Our conclusion here does not necessarily contradict that of Asiedu (2002), which shows that the behavior of FDI to Sub-Saharan Africa differs from that in other regions. We focus here on the southern African countries (SADC), which attracts relatively more FDI than the rest of Sub-Saharan Africa. Moreover, Asiedu (2002) does not control for cross country differences in economic performance.

Although some papers have found significant productivity gaps between foreign-owned and domestic firms, others have reported that this correlation largely disappears after controlling for differences in the quality of inputs. Examples of studies that have found significant productivity premiums for foreign owned firms are Konings (2001), Yasar and Paul (2007), Hallward-Driemeier, Wallsten and Xu (2006), Xu, Zhu and Lin (2005) and Zhang, Zhang and Zhao (2001).

Data

The data set we use to investigate the relationship between foreign ownership and firm performance in this paper comes from the Enterprise Surveys that the World Bank has carried out over the past decade in more than 100 countries. The surveys take a snapshot of a range of indicators of various aspects of a country's business environment, including access to finance and business services, physical and institutional infrastructure, the quality of governance and control of corruption, and information on variables needed to calculate productivity, including revenue from sales and other sources, as well as various expenses. To date, at least one wave of the survey has been conducted in all countries of the SADC except the Seychelles and Zimbabwe, with some countries having had two waves. The member countries covered by the data are: Angola (two waves), Botswana (two waves), Democratic Republic of Congo, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia (two waves), South Africa, Swaziland (two waves), Tanzania (two waves), and Zambia. For each wave and each country, the number of firms in the sample ranges from 75 (Lesotho) to 1057 (second wave of South Africa). Each sample was selected through a stratified random sampling scheme from a sample frame drawn from official listings of the universe of firms for each country. While the timing of the surveys differs across countries, all were carried out between 2005 and 2009.

Table 1 lists the variables used in the firm-level regressions. Table 2 provides descriptive statistics of those variables for the SADC and highlights the differences in sample between foreign owned and domestic firms. A foreign owned firm is one in

which a foreign entity has some ownership share.⁵ A contrast worth noting is that foreign owned firms are significantly larger--the median foreign firm in SADC is 70 log points (or 100 percent) larger than domestic firms. The share of foreign owned firms in the overall sample is also larger in richer countries. Surprisingly, host countries tend to have higher firing costs. A possible explanation is that, in response to opening up to foreign ownership, countries impose higher firing costs to respond to populist demands. Unsurprisingly, foreign firms tend to be located in countries with more competitive banks, which are characterized by lower interest spreads (i.e., the discrepancy between lending and borrowing rates at banks). Similarly, foreign firms tend to be located in countries exhibiting higher indices of government efficiency (by ICRG). Foreign firms tend to suffer lower losses of sales due to crimes.

Our measure of productivity, total factor productivity (TFP hereafter), was estimated from industry-specific translog production functions.⁶ Table 2 suggests that foreign owned firms in the SADC are typically more productive than domestic firms. For example, the median TFP of foreign firms is higher by 5 log points. A foreign owned firm is also more likely to be an exporter than a domestic firm (21 versus 8 percent). In terms of sales growth,⁷ the median growth rate is slightly lower for foreign owned firms than for domestic firms (7.8 versus 9.3 percent), but this reflects the fact the median foreign owned firm is larger than the median domestic firm. That is, the median domestic firm grows faster than the median foreign owned firm because of dynamic economies of scale or learning by doing, but foreign owned firms could grow faster if they were to have the same initial size.

Firm performance and foreign ownership: productivity and growth

In order to identify the sources of the correlation between foreign ownership, productivity and growth of firms, we estimated the following model:

⁵ An alternative definition would be to deem a firm foreign only if foreign entities hold a minimum of ownership share such as 51 percent (or majority) share or some lower threshold. However, it turns out that the quantitative results remain very similar with this alternative definition.

⁶ As a sensitivity check, we also estimate Cobb-Douglas TFP, which is less general than the translog specification. The results are qualitatively very similar.

⁷ Three years of sales are reported in the data.

$$Y_{ic} = FRN_{ic}\delta + F_{ic}\alpha + Z_c\beta + e_{ic} \quad (1)$$

where i refers to a firm, and c refers to a country. Y is either TFP or sales growth. FRN is foreign ownership. The variable set F consists of firm level controls, and includes the share of ownership by the largest shareholder, the age of the firm (in logarithm), and the share of losses of sales due to crime.⁸ The variable set Z consists of country level controls, and include lagged GDP per capita, the ICRG index of government efficiency, the firing costs in terms of years of salary, and the interest spread (as a measure of banking competitiveness). All the country-level variables come from the *World Development Indicators* of the World Bank. Estimations of models with micro and macro units of analysis are known to lead to potential biases in precision, which we addressed by clustering the standard errors at the country level (Moulton 1986).

We report two sets of results, one set with, and the other without, including South Africa in the sample, because South Africa is larger and richer than other African economies. Table 3 reports the results. Since the results without South Africa tend to be very similar, we focus on the first two columns.

The most important finding is that foreign ownership has a strong and significant relationship with both sales growth and TFP. Increasing foreign ownership by 10 percentage points would increase sales growth by 2 percentage points, and TFP by 1.4 percentage points. To put this in perspective, Table A1 in the appendix compares SADC with the East Asia and Pacific region (consisting of China, Indonesia, Malaysia, and Thailand, EAP hereafter) and all non-SADC countries with similar income level as SADC members.⁹ Foreign ownership is significantly higher in SADC (17.5 percent) than in EAP (13.5 percent) and the similar-income sample (10.8 percent). In light of the positive effect of foreign ownership and SADC's advantage in this measure, foreign ownership has reduced the disadvantage of SADC toward other regions.

⁸ We also estimated specifications including the share of losses of sales due to power outages and the average level of schooling of firm employees. The results including these additional controls are similar to the results obtained in the basic specification. However, several countries do not have these indicators. To preserve as many SADC countries as possible in our sample, we opted to report the results without them.

⁹ That is, with income less than South Africa's 4600 U.S. dollars of 2000.

The other findings on performance determinants are also illuminating. The intensity of crime (as measured by the share of sales lost to crimes) is positively associated with sales growth but negatively with TFP. Firms in countries with higher income levels tend to have higher TFP--a country that is richer by 100 log points tends to have a productivity premium of 7.5 percent, which could reflect the forces of a large market or better infrastructure. Unsurprisingly, high firing cost and, thus, a high exit costs, are associated with lower productivity. An interpretation is that firing costs cause shirking by workers and, therefore, reduce the efficiency of a fixed amount of capital and labor. This is consistent with what is commonly found in many developing countries (Xu 2011). Another noteworthy finding is that both sales growth and TFP are negatively affected by high bank interest spreads. Our interpretation is that when banks are less competitive--with a higher interest spread--firms cannot expand as much because capital costs are too high or because it is more difficult to find alternative sources of finance. Similarly, high interest spreads distort capital-labor ratios and, therefore, reduce productivity. Interestingly, countries with higher government efficiency levels are associated with a higher sales growth rates but lower productivity levels. Finally, larger firms tend to grow faster in SADC.

Foreign ownership and exporting status

Exporting may be particularly important for SADC and other small African economies. Exports also signal stronger competitiveness in the region. Thus, one way to gauge the benefits of foreign ownership is to examine whether it also helps firms to export. In general, one expects foreign ownership to facilitate exports because foreign firms tend to have better technologies, management know-how, besides possessing better information on exporting opportunities. On the other hand, foreign owners could enter the local market seeking profits in the domestic market. They may also set up operations in host countries to circumvent import tariffs. To investigate this issue, we thus report the following linear probability model:

$$Export_{ic} = \delta Frn_{ic} + \alpha F_{ic} + \beta Z_c + \varepsilon_{ic}$$

where *Export* is a dummy variable for exporting firms. Subscripts *i* and *c* indicate firm and country. *Frn* is foreign ownership, *F* are firm characteristics, and *Z* are country characteristics. We also experimented with probit and logit specifications, and the results were similar. We report the linear probability model because the coefficients are easier to interpret.¹⁰ Again, standard errors are clustered at the country level. The results are in Table 4. Again, the results with and without including South Africa are quite similar, and we focus on the first column results which include South Africa.

Relative to a domestic firm, the probability of exporting for a foreign firm is higher by almost 6 percentage points. This effect is quite large since the average probability of exporting for domestic firms is only 7.7 percent. Only a few other determinants of exporting status have statistically significant coefficients. Large firms, not surprisingly, tend to export. Interestingly, firms featuring a higher share of ownership by the largest owner (e.g., a proxy of family ownership) are less likely to export.

Are there Spillover Effects of FDI?

Potentially important benefits from FDI are spillovers on domestic firms in terms of transfers of technologies, managerial know-how, and competition effects. These spillover effects can be either horizontal or vertical. Horizontal spillovers refer to the effects of the presence of foreign-owned firms on domestic firms within the same industry (or specializing in the same product). There are two counteracting effects related to horizontal spillovers: (a) the positive aspects such as technological spillovers, personnel movements, transfer of managerial know-how, and the cultivation of more efficient local input suppliers; and (b) the negative aspects associated with foreign firms being competitors of domestic firms, taking away market shares, increasing the average fixed costs, thus reducing the productivity of domestic firms. The positive horizontal effects could be limited by foreign firms' incentives to restrict the flow of commercial secrets to domestic firms, for example, via intellectual property rights and paying for efficiency wages to reduce personnel departures. Indeed, recent studies find insignificant and even negative horizontal spillover effects—sometimes dubbed the

¹⁰ Further controlling for employee schooling and the share of losses due to power shortages does not change our results but reduce the sample significantly.

“market-stealing” effect; see Haddad and Harrison (1993) on Morocco, Aitken and Harrison (1999) on Venezuela, Djankov and Hoekman (2000) on the Czech Republic, and Konings (2001) on Bulgaria, Romania and Poland. The evidence on positive horizontal spillovers appears to be more prominent in developed countries, such as the United Kingdom and the United States (Haskel et al. 2002; Keller and Yeaple 2003). However, Kee (2010) finds positive horizontal FDI spillover effects through local input suppliers used by foreign and domestic garment firms in Bangladesh.

Vertical spillover effects refer to the potential benefits for input suppliers or clients of foreign companies, such as by offering technological help, stronger quality control, and so on. Since there are no conflicts of interests between foreign firms, their clients and input suppliers related to technology and know-how, positive vertical spillover effects are more plausible than positive horizontal spillovers. This is indeed what is found in Javorcik (2004) for Lithuania. Thus, it is safe to say that the evidence on spillovers is mixed. In surveying related evidence, Rodrik (1999) stated that “today’s policy literature is filled with extravagant claims about positive spillover effects from FDI but the evidence is sobering.” Our understanding of spillover effects is further complicated by the idea that the effects of policies tend to differ across contexts due to differences in complementary institutions, regulatory environments, and even skill levels of local employees, which probably shape the nature of inter-firm spillovers (Kremer 1993; Hausman, Rodrik and Velasco 2005; Xu 2011).

The data only allowed us to identify spillover effects within broad industry categories. We first computed country-industry-level average foreign ownership, weighted by the share of employment in the industry ($FRN_{c,j}$). We then estimated the productivity model with the sample of domestic firms only, but added $FRN_{c,j}$ as an additional regressor. A positive coefficient indicates positive spillovers.¹¹ Note that we do not interpret the results as pure horizontal spillovers. The industry category in our data is at the two-digit level. The spillovers effect is thus a mixture of both horizontal and vertical spillovers effects. That is, in addition to horizontal spillovers, our broad externality measure also captures the vertical spillovers of, say, downstream foreign-

¹¹ We dropped firms in industries with fewer than five domestic firms.

owned firms specializing in apparel on upstream domestic firms specializing in yarns or textiles.

The results, in Table 5, show a robust and positive spillover effect from foreign-owned firms. When South Africa is included, $FRN_{c,j}$ has a positive and significant coefficient of around 0.23. The magnitude is nontrivial: increasing average foreign ownership by 10 percentage points would increase domestic firms' productivity in the same industry by 2 percentage points. Thus, besides the direct effect of raising average productivity, foreign-owned firms also seem to enhance the productivity of domestic firms within broad industries. The results, however, strongly hinges on whether South Africa is excluded, in which case, it remains positive, but its magnitude drops to 0.14, and it becomes statistically insignificant (with a t-statistic of 1.51). Thus, the FDI spillover effect is stronger in South Africa than in the smaller southern African countries.

3. SADC's FDI inflows per capita in international perspective

Foreign ownership seems to play a positive role in SADC.¹² In this section we examine the relative performance of SADC countries in attracting FDI, as measured by FDI inflow per capita. In particular, what are the key determinants of FDI? What may explain the differences between SADC countries and comparison groups (such as other countries with similar income levels)? We proceed in several steps. First, we lay out a simple theoretical framework for the key determinants of FDI across countries. Second, we explain our empirical framework and key variables. Finally, we present the results from cross-country regressions on FDI.

A Simple Theoretical Framework

The literature on FDI points towards a relatively simple generic empirical specification (Fan et al., 2009). The literature views prospective multinational firms as possessing

¹² See also Ahmed, Cheng and Messinis (2011), which find that FDI has significant impacts on growth in Sub-Saharan Africa.

information-based firm-specific capabilities that they could profitably apply in foreign countries. Indeed, these capabilities allow them to overcome the “difficulties of being foreign” to generate returns to justify the investment (Morck and Yeung 1991, 1992). Agency problems, information asymmetries, and property rights protection problems that render information-based assets inalienable prevent these firms from selling or leasing those capabilities to foreign firms. To profitably apply their unique capabilities abroad, multinationals resort to establishing controlled foreign operations and engage in FDI. Still, FDI is an investment like any other in the sense that it aims to capture positive net present values.

The net present value of a corporate investment project depends on a number of factors. The first is the size of an economy (Caves 1999). With a larger economy, investment projects with high fixed cost components yield higher net present values due to a larger extent of domestic market that allows the firm to reduce the average cost of the product. All else equal, FDI inflows should rise with the size of the host economy. The net present value a firm sees also depends on local product and factor market development and growth potential. It depends negatively on market risks and the costs of doing business. The latter is perhaps more important and contributing factors include high taxes, high wages relative to productivity, and poor infrastructure.¹³

All these factors hinge on an economy’s institutional environment. If local commercial disputes are adjudicated through transparent and predictable norms or through established legal and political institutions, the disadvantage of being foreign diminishes and FDI flows in. This consideration echoes the finance and growth literature, which emphasizes that sound and well-enforced rules and regulations, like property rights protection and information disclosure, encourage economic development in general and capital market development in particular (La Porta et al. 1997 and 1998; King and Levine

¹³ Coughlin, Terza, and Arromdee (1991) provide empirical support for these factors influencing inward FDI, though they do not consider financial development. Froot and Stein (1991), while showing that undervalued host country currencies attract inward FDI, also stress the barriers firms confront in raising capital to finance new investment projects. These barriers are particularly daunting for domestic firms in economies with underdeveloped capital markets. In such countries, foreign firms could have an advantage in capturing the NPVs of new investment projects because of their access to better functioning foreign capital markets (Foley, Mahir, and Hines, 2004).

1993). The reason is that these rules and regulations constrain opportunistic behavior and build transactional trust between parties (North 1991). Indeed, governments that are less corrupt, have more efficient bureaucracies, and those that impose less burdensome regulations foster economic development (La Porta et al. 1998) and attract FDI (Alfaro et al. 2005, Globerman and Shapiro 2002).

In summary, FDI is attracted by basic economic and institutional factors. The economic factors include the size of the market, the current level of development, and variables such as education and infrastructure development that affect productivity and expected future development. Obviously, other considerations like trade openness and the host country's currency (see, e.g., Froot and Stein 1991) also may affect FDI. The institutional factors are proxied by general indicators of "good government" such as the establishment of law and order as well as high quality government bureaucracy.

The empirical relationship

Our dependent variable is the logarithm of per capita FDI inflows in constant US dollars of 2000, winsorized at the tail 5 percent.¹⁴ We added a constant to the FDI observations because some countries have negative FDI inflows (negative FDI inflow represents repatriation of previous investment). Although dropping the negative FDI inflow observations leads to qualitatively similar results, we do not see a good reason to exclude them.

Our objectives are twofold. First, we want to empirically investigate the correlates of FDI. Second, we want to assess if the determinants of FDI flowing into SADC are similar to those of the FDI flowing into other countries. To these ends, we follow the discussion in the previous section and regress each country's FDI inflows on a set of country characteristics associated with the quality of government, along with some basic measures of the level of development and other country characteristics such as

¹⁴ The winsorization is done to prevent the influence of outliers and measurement errors. This variable is highly skewed. The qualitative results are similar when we winsorize at the 1 or 5 percent levels. However, giving the winsorization threshold is around 0 at 5 percent level, the interpretation of magnitude is a lot easier-- $\log(\text{FDI per capita} + \text{constant})$ is close to $\log(\text{FDI per capita})$ when the constant is 1.4 instead of around 92, and log point is close to percentage point in interpretation.

population size, demographic characteristics and trade policies. We include the SADC dummy in these regressions and then ask whether SADC's FDI inflows behave.

We proxy general institutional quality by a commonly-used indicator of the general quality of government: the rule of law index from ICRG. ***Rule of law*** is an ICRG survey result gauging the state of law and order in each country. It ranges from 1 to 6, with higher values connoting greater general respect for the rule of law.¹⁵ It contains a law component, which captures the strength and impartiality of the legal and political establishment in judicial matters, and an order component, which captures the extent to which residents of a country accept established legal and political institutions as the solely legitimate way to make and implement laws and to adjudicate disputes.

In place of the ICRG corruption index we alternatively use the ***control of corruption*** index. This indicator is most commonly used in the related economics literature. This variable is meant to capture the extent to which illegal payments are expected throughout various levels of government. In addition to being consistent with previous studies, the data has a broad international coverage.¹⁶ The index itself takes on values ranging from zero (most corrupt) to six (least corrupt).

Macroeconomic performance. Macroeconomic performance plays an interesting role. FDI is large if foreign corporate investors regard a location's investment opportunities highly. Obviously, investment opportunities are more abundant in locations with better macroeconomic performance, due perhaps partly to better institutions, where government officials are not corrupt, bureaucracies are efficient, and the rule of law is generally upheld. At the same time, positive shocks on investment opportunities often entice governments seeking to attract foreign capital to provide these institutions.¹⁷ Hence, a simple relationship between measures of government quality and foreign direct investment could be misleading. At the very least, to sort this out, our empirical

¹⁵ ICRG data has the advantage of covering the majority of countries from 1982 on. For details, see Knack and Rahman (2007).

¹⁶ We also considered other indexes such as those from Kaufman-Kraay-Mastruzzi (2010). However, these data start in 1996, which would reduce the estimation sample significantly.

¹⁷ See Easterly et al. (1993), which shows that temporal shocks rather than good policies often lead to positive growth.

investigation should incorporate proxies for the presence of profitable investment opportunities. These are:

Growth trend is a country's per capita GDP growth rate averaged over the previous five years. We interpret a high past growth rate as both indicative of profitable investment opportunities and a track record of the country's government fostering, or at least not impeding, their exploitation.

Macro volatility is the standard deviation of per capita GDP growth over the prior five years. More unstable economic growth, all else equal, is likely less conducive to FDI, and less indicative of sound and predictable government policies.

General development is measured by the following variables: the **log of per capita GDP** in 2000 constant US dollars at PPP; **education**, measured by the log of the average years of schooling completed by the country's adult population; **infrastructure** quality, represented by telephones per 1000 residents; and the **level of urbanization**,¹⁸ which is the share of urban in total population. These are a set of variables commonly used in the literature (see Coughlin et al. 1991).

Following the discussion in the previous section, we incorporate other country characteristics that can be expected to affect FDI. We include **country size**, measured as the log of total population, to control for scale, as well as the the availability of adult labor force, as measured by the **adult share of the population**.¹⁹

We include each country's currency **exchange rate** relative to the US dollar, all normalized by the rate in 2000. This means that a higher value of *exchange rate* implies a more depreciated local currency. Countries with undervalued currencies, all else equal, appear to attract more FDI (e.g., see, Froot and Stein, 1991).

In sensitivity check we also include a measure of **openness** (i.e., imports plus exports divided by GDP) for two reasons. Being open to international trade reduces the

¹⁸ We have also tried including the percent of GDP accounted by manufacturing and services, and found that they don't matter for attracting FDI after controlling for GDP per capita, urbanization, and other controls.

¹⁹ Adults are defined to be between the age of 15 and 64.

need for trade-barrier jumping FDI. The variable is lagged by one period to avoid contemporaneous endogeneity bias.

The SADC dummy. To assess whether FDI inflows to SADC are “exceptional” compared to other countries with similar income levels, we include a SADC dummy variable in the regression.

The specification of our empirical analysis is therefore the following:

$$\ln(FDIPC_{it}) = \alpha_0 + \alpha_{SADC} SADC + X_{it}' \beta + TRACK_{it} \delta + INST_{it} \lambda + \varepsilon_{it} \quad (1)$$

where X represents variables related to general development and other characteristics, including *log GDP per capita*, *log(mean years of schooling)*, *telephone density*, *the urban share of population*, *the adult share of population*, *log population*, *exchange rate*, and *openness*; *TRACK* includes the *growth trend* and its volatility in the past five years; *INST* are the institutional measures represented by *Rule of Law* or *control of corruption*.²⁰ Since many of the explanatory variables, especially the institutional variables, tend to be stable over time, using fixed effects would exacerbate the influence of measurement problems (Griliches and Hausman, 1986), and thus we present OLS estimates. Because the errors are likely to be correlated within countries, we allow clustering of the error term at the country level (Moulton, 1986).

Summary Statistics

Table 6 reports the differences in key variables between the SADC region and the two comparison groups for the 2001 to 2006 period, namely, non-SADC countries and non-SADC countries with similar income levels. The latter group drops country-year observations with incomes exceeding the highest income level for the countries within the SADC region. The SADC region has an average growth rate in real GDP per capita

²⁰ There are some missing observations for rule of law, corruption, and schooling. Since dropping all missing observations for these variables would entail a significant loss of sample, we imputed these variables with basic country indicators such as GDP per capita and the urbanization share; we then also include missing indicators for the three variables to capture potential mis-imputation. We reach similar conclusions about our key results if we drop the sample with imputed missing observations for the three variables.

of 2.3 percent, lower than the non-SADC group (3.1 percent) and the similar-income sample (3.1 percent) by 0.8 percentage points. The average GDP per capita is around 1300 U.S. dollars, slightly higher than the similar-income sample, and much lower than the non-SADC sample. Per capital FDI is 36.6 U.S. dollars (in 2000 value), about 18 percent of the average for the non-SADC sample (202.8 dollars), and 58 percent of the similar-income sample (63.2 dollars). The two proxies of institutions, the rule of law index and the control of corruption index, are similar between SADC and the similar-income sample, but slightly lower than the non-SADC sample. Thus, it appears that SADC lags slightly behind in attracting FDI inflows relative to similar-income countries and further behind when compared with the non-SADC sample. Moreover, it seems that the institutional disadvantage is negligible; key disadvantages for SADC seem to be its lower growth rate and its lower income level to the extent that these factors are important.

Results

Table 7 reports the results. The pooled-sample results appear in the first three columns, and the similar-income sample results are reported in the next two columns. The institutional variable is the rule of law in Column (1), and the control-of-corruption index in Column (2). Column (3) adds "trade/GDP lagged" as an additional control to test the robustness of our results. The structure of the last two columns with the similar-income subsample is similar, but for the sake of brevity we do not report the results using the control-of-corruption index as an alternative institutional measure.²¹

The alternative measures of institutions yield similar results (see columns (2) and (3)). In addition, adding lagged trade/GDP does not change our main results. However, since trade/GDP can be affected by FDI, it is potentially endogenous. Thus, we leave it out of our base specification. But for readers who are concerned that a major omitted variable might be openness, our main results are unaffected by its inclusion. Finally, the main results from the similar-income sample are also similar to those of the pooled sample.

²¹ The results are very similar and are available upon request.

We thus focus on Column (1). Good macroeconomic performance track records attract FDI. The average growth in the previous five years appears with a positive and significant coefficient while growth volatility has the opposite sign (though statistically insignificant). The general government quality, the *rule of law* variable, does not directly affect FDI. Growth expectations, therefore, are the most important determinant of FDI.

Measures of economic fundamentals matter a great deal. Schooling, though statistically insignificant, has the expected sign. Phone density is significant, indicating that FDI tends to flow to countries with better infrastructure. Similarly, FDI tends to flow to richer countries, due perhaps to better protection of property rights, market sizes and so on. Another important variable is the share of adult labor force in total population, which has the expected positive sign and is highly significant. This is potentially worrisome for SADC because the region has a higher share of dependents than other regions. Interestingly, countries with a smaller population tend to have higher FDI per capita—perhaps the benefits of FDI (such as through bringing export opportunities) are higher for smaller countries, which face a tighter constraint of small domestic markets.

The SADC dummy is not statistically significant. This means that once we control for the observed variables, the region on average does not lag behind other countries in attracting FDI per capita. It appears that SADC's relatively low FDI is therefore a consequence of relative under-performance in terms of the observed explanatory variables.

A more general test of the SADC effect is to allow SADC to interact with the other explanatory variables. It turns out that only the interaction terms with phone density, income level, and openness are close to being statistically significant. Thus, we report a specification in Table 8 in which the SADC dummy variable is interacted with these three variables. The coefficients of the non-interacting variables remain largely the same as those reported in Table 7. The SADC dummy variable still is statistically insignificant. The three interaction terms, $SADC \times GPD$ per capita, $SADC \times openness$, and $SADC \times phone\ density$ are all statistically significant. For SADC, phone density is not important for attracting FDI. Neither is the income level, which makes sense because

for the relatively small SADC countries, market size (partially captured by the level of income) is not important because FDI probably tends to be export-oriented, a tendency confirmed by our earlier firm-level analysis. Finally, openness is much more important, with a coefficient that is twice that of the rest of the world. Again, our interpretation is that for small countries, openness increases the expected value of FDI as openness facilitates exports. In other words, there is a stronger complementarity of openness and FDI for SADC.

What Explains the SADC Disadvantage in FDI?

To shed light on how the key variables explain the relatively low FDI per capita of SADC, we choose two comparison groups: the East-Asia Pacific countries (EAP, consisting of China, Indonesia, Malaysia, and Thailand), and the similar-income sample (consisting of all countries in our sample that have the same range of income per capita as the SADC). While our choice of the similar-income sample is easy to understand, we chose EAP to explain the gap between SADC and the front-runners among the developing countries. The results are contained in Table 9.

The first comparison concerns SADC versus EAP. The difference in the dependent variable is 1.08 in favor of EAP. The percent of this difference explained by variable X_k is calculated as $100 * \beta_k (EX_{k,EAP} - EX_{k,SADC}) / \Delta Y$, where $\beta_k, EX_{k,R}, \Delta Y$ are the estimated coefficients (with the specification under the first column of Table 7 without the SADC dummy), the mean X_k for region R , and the difference in the outcome, respectively. The higher adult share of population in EAP explains 43 percent of the observed difference. The higher previous growth rate explains a further 32 percent. EAP's advantage in phone density explains another 24 percent. The income level advantage of EAP adds 16 percent, and a better schooling becomes the last (somewhat) important factor, explaining about 9 percent.²² Country sizes, which is negatively correlated with FDI, actually help SADC countries since they tend to be smaller.

²² The sum of the shares of the contribution of reported explanatory variables need not add to 100 percent because we do not report the shares that are explained by other variables, and because SADC and the comparator countries do not constitute the full sample of the regression.

The second comparison is between SADC and countries with similar income. The comparison group has an advantage in the dependent variable of 0.41. The most important variables for explaining this difference are phone density (43 percent of the difference), the adult share of population (39 percent), the income level (23 percent), and previous growth rates (7 percent).

4. Conclusions

At the beginning we asked three questions about FDI in SADC: Have FDI inflows played a positive role in SADC economies? Are SADC countries attracting too little FDI? What can be done to attract more FDI to this region? The answer to the first question is yes. We found evidence that FDI helped southern African economies. Foreign firms tend to be larger, export more, have higher productivity and sales growth rates; the presence of these firms thus raises the average firm performance and creates jobs for host countries. Moreover, domestic firms in the same industry tend to benefit from a stronger presence of foreign firms.

Regarding the second question, we found that FDI flows to countries with good macroeconomic performance, as indicated by stable high growth in the past and good economic fundamentals (i.e., phone density, the adult share of population, and income level). Usual suspects – corruption or the rule of law – do not appear to significantly affect FDI. Importantly, we did not find that SADC's ability to attract foreign direct investment is exceptional: the SADC dummy is never statistically different from zero. There may be some differences between SADC and the rest of the world in FDI behavior, however: the income level appears less important, while openness to international trade is particularly important.

The comparisons with other regions suggest that SADC has lower FDI per capita due to poorer economic fundamental (previous growth rates, lower income level, phone density, and the adult share of the population). The accounting exercise raises another question: Why do SADC countries have worse economic fundamentals? This is beyond

the scope of this paper. But there are plenty of analyses shedding light on this question. Some of the potential culprits include ethnic polarization (Easterly and Levine, 1997), political instability and civil wars (Collier and Gunning 1999), among others. The key for attracting FDI in SADC would be to build a fundamentally sound economy that naturally lures FDI. In addition, the results indicate that openness is especially important for SADC. Policies and procedures to encourage openness would therefore be especially important for the region.

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Table 1. Definitions of variables in firm-level analysis

Variable	Definitions
FRN	Share of foreign ownership
largest owner	Ownership shares by the largest shareholder.
ln(firm age)	The logarithm of firm age.
Loss_crime	The share of losses in sales due to crime.
Ln(GDPpc)	Ln(GDP per capita of the country)
Firing_cost	Firing costs in terms of weeks of wages, from WDI.
Interest_spread	Interest spread, from WDI.
Govt_efficiency	Government effectiveness index, from ICRG.
Control_CR	Control of corruption, from ICRG.

Table 2. Firm-level summary statistics for the SADC region as a whole

	Foreign firms			Domestic firms		
	mean	p50	sd	mean	p50	sd
Ln(L)	3.395	3.091	1.619	2.530	2.398	1.333
Ln(GDPpc)	7.013	7.230	1.201	6.852	6.951	1.181
Firing_cost	61.230	53.200	41.204	53.537	32.000	41.339
Interest_spead	10.583	7.587	10.571	12.157	8.720	12.880
Govt_efficiency	-0.176	-0.306	0.746	-0.273	-0.334	0.754
Sales growth	0.405	0.078	1.370	0.285	0.093	1.008
TFPtran	0.091	0.023	0.581	0.001	-0.030	0.492
Exporter	0.207	0.000	0.406	0.077	0.000	0.267
FRN	0.834	1.000	0.264	0.000	0.000	0.000
Largest owner	0.711	0.700	0.245	0.839	1.000	0.241
Ln(firm age)	2.139	2.079	0.984	2.036	1.946	0.968
Loss_crime	0.012	0.000	0.038	0.021	0.000	0.514

Table 3. Effects of foreign ownership: SADC sample

	All SADC countries		Excluding South Africa	
	sales growth	TFPtran	sales growth	TFPtran
	coef/se	coef/se	coef/se	coef/se
foreign ownership	0.200** (0.081)	0.141*** (0.048)	0.151* (0.084)	0.097** (0.045)
largest owner	0.174** (0.075)	0.029 (0.026)	0.215** (0.085)	0.037 (0.032)
ln(firm age)	-0.168*** (0.049)	-0.018 (0.012)	-0.190*** (0.050)	-0.016 (0.014)
loss_crime	0.039*** (0.003)	-0.022*** (0.002)	0.040*** (0.003)	-0.021*** (0.001)
ln(GDPpc)	-0.218*** (0.053)	0.075** (0.036)	-0.199*** (0.045)	0.085** (0.036)
Firing_cost	-0.001* (0.001)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
interest spread	-0.015*** (0.003)	-0.004*** (0.001)	-0.015*** (0.003)	-0.004*** (0.002)
government effectiveness	0.169* (0.101)	-0.124* (0.064)	0.217** (0.093)	-0.082 (0.062)
lnS	0.041 (0.025)		0.065*** (0.023)	
industry dummies	yes	no	yes	no
Number of observations	3,842	2,929	3,186	2,299
Adjusted R2	0.094	0.066	0.105	0.078

standard errors in parentheses, clustered at the country level.

*, ** and ***: statistical significance at the 10, 5 and 1 percent levels.

**Table 4. Exporting and Foreign ownership:
The SADC sample**

	All SADC countries	Excluding South Africa
	exporter	exporter
	coef/se	coef/se
foreign	0.055*** (0.021)	0.050*** (0.019)
largest owner	-0.061*** (0.024)	-0.050** (0.023)
ln(firm age)	0.012 (0.009)	0.005 (0.006)
loss_crime	0.000 (0.001)	0.000 (0.001)
Lgdppc	0.000 (0.016)	0.002 (0.018)
Firing_cost	-0.000 (0.000)	-0.000 (0.000)
interest spread	-0.001 (0.001)	-0.001 (0.001)
government effectiveness	-0.008 (0.021)	0.000 (0.024)
lnS	0.037*** (0.004)	0.038*** (0.005)
industry dummies	yes	yes
Number of observations	6,248	5,368
Adjusted R2	0.189	0.197

standard errors in parentheses, clustered at the country level.

*, ** and ***: statistical significance at the 10, 5 and 1 percent levels.

Table 5. Effects of foreign ownership on productivity of SADC domestic firms

	All SADC countries		Excluding South Africa	
	TFPtran	TFPtran	TFPtran	TFPtran
	coef/se	coef/se	coef/se	coef/se
largest owner	0.070** (0.028)	0.066** (0.026)	0.091*** (0.028)	0.090*** (0.027)
ln(firm age)	-0.021 (0.014)	-0.023 (0.015)	-0.018 (0.016)	-0.019 (0.016)
loss_crime	-0.021*** (0.001)	-0.022*** (0.001)	-0.021*** (0.001)	-0.021*** (0.001)
ln(GDPpc)	0.094*** (0.026)	0.081*** (0.029)	0.107*** (0.028)	0.099*** (0.027)
firing_cost	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
interest spread	-0.003*** (0.001)	-0.003** (0.001)	-0.003*** (0.001)	-0.003** (0.001)
government effectiveness	-0.148*** (0.057)	-0.130* (0.067)	-0.079 (0.057)	-0.065 (0.058)
industry-level employment-weighted average foreign ownership	0.232*** (0.082)		0.139 (0.092)	
Number of observations	2,210	2,210	1,669	1,669
Adjusted R2	0.066	0.062	0.087	0.086

Note. Standard errors in parentheses, clustered at the country level.

***, **, and * represent statistical significance at the 1, 5 and 10 percent levels.

The sample consists of domestic firms in SADC countries..

**Table 6. Summary Statistics for SADC and Comparator countries:
2001-2006 average.**

	SADC	Non-SADC	Non-SADC: similar income countries
	Mean	Mean	Mean
Real growth rate of GDP per capita	2.261	3.128	3.102
GDP per capital, USD real	1,289.358	7,248.203	1,229.144
FDI per capita	36.599	202.808	63.167
FDI/GDP	0.037	0.068	0.041
Rulelaw	0.991	1.181	0.946
Control_CR	0.651	0.834	0.661

Table 7. Is the SADC region different?
The dependent variable is ln(FDI per capita + constant)

	Pooled Sample			Similar-Income Sample	
	coef/se	coef/se	coef/se	coef/se	coef/se
ln(schooling _{t-1})	0.198 (0.170)	0.200 (0.166)	0.138 (0.177)	0.218 (0.169)	0.065 (0.175)
ln(phone density _{t-1})	0.243** (0.116)	0.231** (0.111)	0.261** (0.114)	0.159 (0.110)	0.121 (0.106)
the adult share of population _{t-1}	4.391** (2.103)	4.456** (2.086)	2.505 (2.097)	4.914** (2.080)	3.621* (2.000)
urban share _{t-1}	0.334 (0.614)	0.348 (0.610)	0.145 (0.616)	0.378 (0.653)	0.456 (0.642)
Ln(population)	-0.123* (0.063)	-0.124** (0.063)	-0.009 (0.068)	-0.177*** (0.061)	-0.047 (0.064)
Relative exchange rate _{t-1}	-0.002* (0.001)	-0.002 (0.001)	-0.002** (0.001)	0.003 (0.008)	0.001 (0.009)
Ln(GDPpc _{t-1})	0.275* (0.148)	0.268* (0.149)	0.326** (0.151)	0.300* (0.159)	0.376** (0.156)
Mean GDPpc growth in the previous 5 yrs	0.075*** (0.015)	0.074*** (0.015)	0.072*** (0.013)	0.069*** (0.015)	0.068*** (0.014)
S.D. GDPpc growth in the previous 5 yrs	-0.890 (0.714)	-0.860 (0.706)	-1.304* (0.738)	-0.693 (0.680)	-1.071 (0.715)
Rulelaw _{t-1}	0.025 (0.056)		0.036 (0.055)	0.018 (0.057)	0.014 (0.058)
Control of Corruption _{t-1}		0.064 (0.070)			
Trade/GDP _{t-1}			0.702*** (0.160)		0.960*** (0.195)
SADC dummy	0.018 (0.305)	-0.011 (0.306)	-0.117 (0.286)	0.049 (0.287)	-0.085 (0.255)
Number of observations	3,265	3,265	3,187	2,311	2,269
Adjusted R2	0.442	0.443	0.455	0.376	0.409

Note. ***, **, and * represent statistical significance at the 1, 5 and 10 percent levels.
Standard errors clustered at the country level.
The constant in the dependent variable is 1.4.

Table 8. Is SADC Different? Another Look
The dependent variable is ln(FDI per capita + constant)

ln(schooling _{t-1})	0.109 (0.631)
ln(phone density _{t-1})	0.308*** (2.711)
the adult share of population _{t-1}	2.560 (1.263)
urban share _{t-1}	0.091 (0.148)
Ln(population)	-0.015 (-0.228)
Relative exchange rate _{t-1}	-0.002* (-1.743)
Ln(GDPpc _{t-1})	0.329** (2.149)
Mean GDPpc growth in the previous 5 yrs	0.077*** (5.839)
S.D. GDPpc growth in the previous 5 yrs	-1.388* (-1.862)
Rulelaw _{t-1}	0.018 (0.341)
Trade/GDP _{t-1}	0.639*** (3.924)
SADC	1.928 (1.347)
SADC * ln(phone density _{t-1})	-0.293* (-1.791)
SADC * Ln(GDPpc _{t-1})	-0.386* (-1.696)
SADC * Trade/GDP _{t-1}	0.683* (1.939)
Number of observations	3,187
Adjusted R2	0.464

Note: ***, **, and * represent statistical significance at the 1, 5 and 10 percent levels.

Standard errors clustered at the country level.

The constant in the dependent variable is 1.4.

Table 9. Accounting for the log FDI per capita outcome

	Coeff	Mean for		$\Delta y = 1.08$ $\beta_k (EX_{EAP} - EX_{SADC}) / \Delta Y$
Independent variables:		SDDC 2.09	EAP 3.17	
Share of adult population _{t-1}	4.37	0.53	0.64	42.58
Mean real GDPpc growth rate in the previous five years	0.08	0.59	5.17	31.81
ln(phone density _{t-1})	0.24	0.42	1.50	24.23
ln(GDPpc _{t-1})	0.27	6.41	7.05	16.35
ln(schooling)	0.20	1.10	1.60	9.33
urban _{t-1}	0.33	0.31	0.37	2.08
S.D. of real GDPpc growth rate in the previous five years	-0.89	0.07	0.06	1.32
rulelaw _{t-1}	0.03	2.62	3.14	1.22
Relative exchange rate _{t-1}	0.00	2.71	0.69	0.32
ln(population)	-0.12	1.91	4.83	-33.30
		similar income countries: mean 2.50		$\Delta y = 0.41$
		SDDC mean 2.09		
ln(phone density _{t-1})	0.24	0.42	1.15	43.04
Share of adult population _{t-1}	4.37	0.53	0.57	38.51
ln(GDPpc _{t-1})	0.27	6.41	6.75	22.66
urban _{t-1}	0.33	0.31	0.43	9.64
Mean real GDPpc growth rate in the previous five years	0.08	0.59	0.98	7.09
ln(population)	-0.12	1.91	1.66	7.58
ln(schooling)	0.20	1.10	1.18	3.86
Relative exchange rate _{t-1}	0.00	2.71	0.69	0.84
S.D. of real GDPpc growth rate in the previous five years	-0.89	0.07	0.08	-2.53
rulelaw _{t-1}	0.03	2.62	2.46	-1.01

Notes: The results for other variables are not reported. The coefficients are based on the same specification in Column (1) of Table 7--except that we do not control for SADC dummy here, since it is not statistically significant.

The Appendix

Table A.1. Mean Characteristics for SADC, EAP and the similar-income sample

	mean	mean	mean
	SADC	EAP	Similar income as SADC
Foreign	0.175	0.135	0.108
Largest owner	0.813	0.678	0.753
ln(firm age)	2.058	2.354	2.324
Loss_crime	0.020	0.004	0.012
ln(GDPpc)	6.886	7.247	6.937
Fire_cost	55.152	88.344	64.715
interest_spread	11.822	3.542	9.335
govt_efficiency	-0.253	0.052	-0.275
ln(sales)	4.856	8.288	6.361

Note. “Similar income as SADC” means income less than 4600 USD (in 2000 value)